

IVANO-FRANKIVSK, Ukraine

WMO: 335260

Lat: 48.8892N

Lon: 24.6892E

Elev: 280

StdP: 98.01

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.4	-15.2	-21.6	0.6	-19.0	-17.4	0.8	-14.8	11.0	3.4	10.0	3.1	0.7	300	0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	30.8	20.4	28.9	19.9	27.1	19.1	21.8	27.9	20.9	26.8	20.0	25.5	3.1	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.9	15.1	24.9	18.9	14.2	23.5	18.0	13.4	22.5	65.1	28.2	61.4	26.8	58.3	25.8	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	7.9	6.9	DB	-24.1	33.8	4.9	1.9	-27.7	35.1	-30.5	36.2	-33.3	37.2	-36.9	38.6	(4)
(5)			WB	-24.3	23.7	4.9	1.5	-27.8	24.8	-30.6	25.7	-33.4	26.5	-36.9	27.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.6	-3.4	-1.4	2.9	9.3	14.3	17.7	19.7	19.0	14.1	8.6	3.4	-1.8		
	DBStd	9.20	5.83	6.04	4.89	4.01	3.48	3.13	2.87	2.94	3.40	4.13	4.47	5.63		
	HDD10.0	1672	416	319	222	59	6	0	0	0	6	77	202	366		
	HDD18.3	3711	674	552	478	273	131	49	19	27	132	303	448	624		
	CDD10.0	1152	0	0	2	37	139	230	299	280	128	33	4	0		
	CDD18.3	149	0	0	0	0	6	29	60	49	6	0	0	0		
	CDH23.3	1693	0	0	0	19	115	315	594	540	103	6	0	0		
	CDH26.7	477	0	0	0	3	16	67	184	183	23	0	0	0		
(14) Wind	WSAvg	2.6	2.5	2.8	3.2	3.0	2.6	2.5	2.4	2.1	2.2	2.3	2.5	2.5		
(15) Precipitation	PrecAvg	691	29	32	41	50	82	97	100	75	67	52	34	33		
	PrecMax	1017	50	81	105	102	169	237	261	158	133	128	66	68		
	PrecMin	497	9	11	12	16	45	42	17	14	17	5	0	7		
	PrecStd	135	10	13	24	19	31	46	55	36	32	32	19	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	14.1	19.0	26.0	28.4	30.5	33.2	33.1	29.2	24.1	17.9	12.2		
		MCWB	7.1	8.2	10.2	15.3	17.5	20.2	21.3	20.8	18.8	15.9	11.9	7.9		
	2%	DB	7.9	10.8	15.4	22.1	26.2	28.6	30.8	30.9	26.1	20.9	14.8	9.0		
		MCWB	5.1	6.6	8.9	13.0	16.9	19.9	20.8	20.3	17.5	14.6	10.8	6.2		
	5%	DB	5.8	8.1	12.8	19.8	24.2	26.9	28.8	28.8	23.8	18.0	12.1	6.2		
		MCWB	3.5	5.2	7.5	12.0	16.5	19.2	20.3	19.8	16.8	13.2	9.1	4.3		
	10%	DB	3.9	6.0	10.2	17.2	22.1	25.0	26.8	26.2	21.1	15.3	10.0	4.6		
		MCWB	2.1	3.8	6.4	10.7	15.5	18.4	19.5	19.1	15.8	12.0	8.3	3.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	8.9	11.0	15.6	19.4	22.4	23.0	23.2	20.0	16.8	12.6	8.3		
		MCDB	9.9	13.3	17.4	24.3	25.5	28.1	29.8	30.2	26.9	22.2	16.6	11.5		
	2%	WB	5.3	7.0	9.5	13.9	18.2	20.9	21.9	21.7	18.6	15.3	11.1	6.2		
		MCDB	7.5	10.0	14.4	20.5	24.1	26.3	28.2	27.7	24.5	19.9	14.2	8.4		
	5%	WB	3.7	5.5	8.1	12.7	17.1	19.9	21.0	20.6	17.3	13.8	9.7	4.6		
		MCDB	5.3	7.8	11.8	18.3	22.9	25.2	27.2	26.5	22.1	17.3	11.7	6.0		
	10%	WB	2.2	4.0	6.7	11.4	16.0	18.9	20.0	19.6	16.1	12.3	8.2	3.2		
		MCDB	3.6	5.8	9.8	16.3	21.0	23.9	25.5	25.0	20.4	15.0	9.9	4.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.0	7.9	9.2	11.4	11.7	11.0	11.2	11.8	10.9	10.1	7.3	6.8		
		MCDBR	8.8	11.1	14.7	16.9	16.2	15.0	15.4	16.8	16.6	15.4	11.8	10.0		
	5% WB	MCWBR	6.5	7.5	8.8	8.6	8.1	7.4	6.8	7.4	8.6	9.5	8.1	7.6		
		MCWBR	8.3	10.2	12.5	14.8	14.1	12.9	13.5	14.3	14.4	13.6	10.7	9.0		
(40) Clear-Sky Solar Irradiance	taub		0.307	0.336	0.379	0.432	0.416	0.415	0.441	0.443	0.399	0.364	0.336	0.305		
	taud		2.291	2.246	2.210	2.131	2.244	2.273	2.213	2.206	2.286	2.360	2.380	2.356		
	Ebn at Noon		738	800	822	813	846	847	817	795	794	761	702	694		
	Edh at Noon		73	96	118	142	132	129	135	130	109	86	68	61		
(44) All-Sky Solar Radiation	RadAvg		1.02	1.82	2.90	4.16	5.16	5.48	5.37	4.85	3.41	2.17	1.07	0.75		
	RadStd		0.14	0.24	0.37	0.43	0.52	0.41	0.45	0.45	0.52	0.31	0.13	0.11		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.73	N/A	N/A	+1.31	+0.60	+0.64	N/A	-304	+130	+48
(47) Regional (21 neighbors)		+0.68	N/A	N/A	+1.03	+0.45	+0.45	-132	-203	+118	+50

Nomenclature: See separate page